

Clinical Impact of the Refit CKD-EPI 2021 Creatinine-Based eGFR Equation

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BACKGROUND: The National Kidney Foundation recently endorsed the refit Chronic Kidney Disease Collaboration (CKD-EPI) equation for estimated glomerular filtration rate (eGFR) using creatinine, age and sex [2021 eGFR_{Cr}(AS)] without a coefficient for race. We evaluated the impact of adopting the 2021 eGFR_{Cr}(AS) equation or a variation of the 2009 CKD-EPI eGFR equation without race [2009 CKD-EPI eGFR_{Cr}(ASR-NB)] compared to the original CKD-EPI eGFR [2009 eGFR_{Cr}(ASR)].

METHODS: The studied population included patients with a clinically ordered iothalamate clearance ($n = 33\,889$). Bias was assessed as the difference between measured and estimated GFR. P30 was defined as the percentage of estimates within 30% of measured GFR, and concordance was determined according to relevant clinical thresholds.

RESULTS: Among Black patients, the median bias for 2009 eGFR_{Cr}(ASR), 2009 eGFR_{Cr}(ASR-NB), and 2021 eGFR_{Cr}(AS) was $-1.32\text{ mL min}^{-1}(1.73\text{ m}^2)^{-1}$ (95CI -2.46 to -0.26), $-8.81\text{ mL min}^{-1}(1.73\text{ m}^2)^{-1}$ (95CI -9.93 to -7.58), and $-6.08\text{ mL min}^{-1}(1.73\text{ m}^2)^{-1}$ (95CI -7.18 to -4.92), respectively. The median bias among non-Black patients was $-0.15\text{ mL min}^{-1}(1.73\text{ m}^2)^{-1}$ (95CI -0.84 to -0.08) for 2021 eGFR_{Cr}(AS) compared to $-3.09\text{ mL min}^{-1}(1.73\text{ m}^2)^{-1}$ (95CI -3.17 to -3.03) for the 2009 eGFR_{Cr}(ASR). P30 and concordance were not significantly different in either racial group. The net reclassification improvement at a measured GFR $<20\text{ mL min}^{-1}(1.73\text{ m}^2)^{-1}$ was 6.4% (95CI 0.36 to 12.4) for Black patients and -5.1% (95CI -6.0 to -4.1) for non-Black patients using the 2021 eGFR_{Cr}(AS) equation.

CONCLUSIONS: Overall, the change in reported eGFR was minimal. However, these changes led to significant reclassification improvements at lower eGFR, which will indirectly improve equitable access to CKD resources.

Introduction

In 1999, the medical community was introduced to the Modification of Diet in Renal Disease Study serum creatinine-based equation, which employed several demographic variables to predict an estimated glomerular filtration rate (eGFR) from serum creatinine (1). The study team empirically observed that race (self-reported Black or non-Black) was a significant variable in the regression models for eGFR. The data suggested mean serum creatinine was higher among Black people compared to non-Black people with similar GFR, perhaps due to increased creatinine generation from muscle. However, the study did not directly measure muscle mass, and this assumption relied on older smaller studies (2–5).

This paradigm was carried on when race was included as a variable in the 2009 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation (6). Although the available study data supported inclusion of a race coefficient in both equations, this practice also implicitly suggested that Black patients were biologically different from other patients, and that perhaps their kidneys worked differently. Beyond the sociological impacts and confusion regarding categorization of biracial persons, the analytical uncertainty of calculated eGFR is significantly larger than the 16% adjustment attributed to race (7).

The national debate regarding race-based medicine has highlighted that the use of race in clinical algorithms (such as eGFR) can be highly problematic, especially given the history of systemic racism in medicine (2–5). Further, it is well accepted that race is a social construct rather than a biological construct (8, 9). Inclusion of race in medical decision making oversimplifies the true diversity within and among patient populations. Singling out a specific group, as the 2009 CKD-EPI eGFR_{Cr} and 2012 CKD-EPI eGFR_{Cr-cys} equations did,

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has the potential to arbitrarily burden a targeted population with systematic healthcare consequences (10).

Clinical laboratory reporting of National Kidney Foundation recommended creatinine-based eGFR has played a role in raising awareness of CKD in the USA (11–14). In addition to increased awareness of CKD, access to standardized creatinine methods and eGFR equations has led to the development and implementation of many care metrics and medical decision points that rely on specific eGFR thresholds (15). Unfortunately, the fixed decision values do not account for the uncertainty in calculated eGFR and imprecision of gold-standard iothalamate clearance (7, 16). Nevertheless, the threshold most often used to define CKD is a GFR $<60 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$. Antibiotics, radiology contrast agents, chemotherapies, and many other pharmaceutical agents have well-defined GFR thresholds below which their dose should be adjusted and/or they should not be used. Similarly, to be listed for a deceased-donor kidney transplant, patients typically must have a measured or estimated GFR $<20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$. Despite these efforts including use of the current race-based eGFR equations, CKD remains underdiagnosed in Black patients, who also experience lower referrals to nephrology specialists and lower rates of transplant listing (17).

Given these concerns, new refit eGFR equations without race coefficients were recently developed and published by the CKD-EPI Collaboration (18). The new equations have been endorsed by the National Kidney Foundation and the American Society of Nephrology (19). Rapid implementation of this method has been encouraged; however, after over a decade of widespread use of the current eGFR equations that contain a race coefficient, it is important to understand the impact this new equation might have on clinical decision making in practice. Thus, in this report we compared the 2009 and 2021 CKD-EPI eGFR_{cr} equations in a large patient cohort with measured GFR (mGFR) by renal iothalamate clearance performed for clinical care.

Methods

PATIENT POPULATION

All patient data was accessed in compliance with the Mayo Clinic Institutional Review Board. Laboratory results were abstracted from the medical record for stable outpatients with a clinically ordered GFR measured by iothalamate renal clearance between October 2006 and September 2021 at Mayo Clinic, Rochester, MN, and a concurrent (within 3 d of mGFR) serum creatinine measured by a standardized enzymatic creatinine assay (implemented in the Mayo Clinic Central Chemistry Laboratory in 2006). Patients who

underwent renal function measurement for chemotherapy dosing, were paraplegic, quadriplegic, <18 years of age, amputees, or kidney transplant recipients were excluded. The remaining 25 512 patients with 33 889 mGFR measurements were included in the study cohort. Race was documented by questionnaire, with all patients that self-identified as Black, African, or African-American grouped as Black, while all others were grouped as non-Black.

IOTHALAMATE CLEARANCE, SERUM CREATININE, AND EGFR

GFR was measured by nonradiolabeled iothalamate clearance. Patients were asked to report for testing early in the day and fasting to minimize diurnal or dietary variations. Iothalamate was administered subcutaneous followed by oral hydration to encourage urine flow. Plasma and urine were collected in timed intervals for iothalamate quantification by liquid chromatography-tandem mass spectrometry in the Mayo Clinic Renal Testing Laboratory (20). Measured GFR was corrected for body surface area as estimated by the DuBois formula and normalized to 1.73 m^2 (21). Serum creatinine was measured using a Roche Cobas chemistry analyzer (c701 or c501; Roche Diagnostics; Indianapolis, USA) running enzymatic methods standardized to international reference materials in the Mayo Clinic Central Clinical Chemistry Laboratory (22). GFR was estimated using 3 different equations: the 2009 CKD-EPI eGFR_{cr} equation that includes age, sex, race [2009 eGFR_{cr}(ASR)]; a variation of the 2009 CKD-EPI eGFR_{cr} equation that omits the “1.159 if Black” coefficient [2009 eGFR_{cr}(ASR-NB)]; and the 2021 CKD-EPI eGFR_{cr} for all races that only includes age and sex terms [2021 eGFR_{cr}(AS)] (18).

STATISTICAL ANALYSIS

Equation performance was compared between the 3 different equations within each race group. Bias was calculated as the median difference between eGFR and mGFR and significance was determined by Wilcoxon rank-sum. P30 was defined as the percentage of estimates $<30\%$ different from measured GFR. Concordance (% agreement) between eGFR and mGFR was determined for clinically defined categories of <15 , 15–29, 30–44, 45–59, 60–89 and $\geq 90 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$. Net reclassification improvement was determined by first binning patients according to mGFR at the stated clinical decision point. Next, the number of patients reclassified into the same category as mGFR by the 2021 eGFR_{cr}(AS) or 2009 eGFR_{cr}(ASR-NB) compared to the 2009 eGFR_{cr}(ASR) was determined. Confidence intervals of P30 and agreement categories were determined by the Wald method and significance was determined by *t*-test. All statistical

analyses were performed using JMP software (SAS Institute, Cary, NC).

Results

ACCURACY OF EGFR AS COMPARED WITH MGFR

Black patients accounted for 852 (2.5%) samples. There were no significant differences in height, weight, or body-mass index (BMI) by race (Table 1). Patients were between 40–65 years in 62% of Black patients and 60% of non-Black patients. The mean $\pm$ SD for both mGFR [70.5 ± 35.2 vs 67.2 ± 32 mL min⁻¹ (1.73 m²)⁻¹] and serum creatinine (1.69 ± 1.44 vs 1.40 ± 0.90 mg/dL) were higher for Black patients. Measured GFR was plotted as a function of eGFR for each equation and patient group (Fig. 1). Root mean square error and slope were similar for all 3 equations within each group.

All 3 eGFR equations underestimated mGFR in both racial groups (Table 2). The mean difference between eGFR and mGFR among non-Black patients was significantly smaller for 2021 eGFR_{Cr}(AS) compared to 2009 eGFR_{Cr}(ASR). Among Black patients, the mean

bias was smallest for the 2009 eGFR_{Cr}(ASR) equation. Compared to 2009 eGFR_{Cr}(ASR), P30 was significantly lower for 2009 eGFR_{Cr}(ASR-NB) among Black patients (64.9% vs 71.7%; $P = 0.0025$) while the 2021 eGFR_{Cr}(AS) not significantly different (71.7% vs 68.7%; $P = 0.17$). P30 and categorical agreement were similar for both equations among non-Black patients. Bias was inversely related to BMI with eGFR overestimating mGFR among patients with BMI <25 kg/m² for all equations and groups. No significant relationships with age were observed for any equation or patient group.

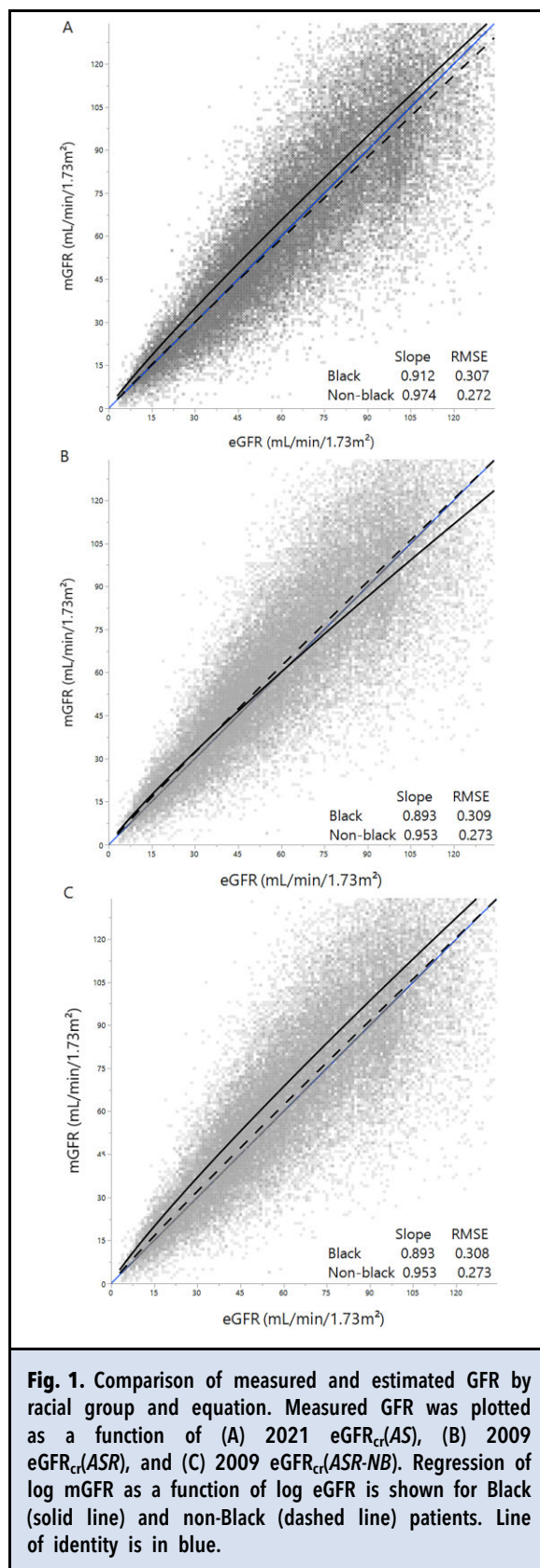
RECLASSIFICATION IMPROVEMENT

The 2009 eGFR_{Cr}(ASR) equation concordantly classified 82% (281 of 342) of Black and 88% (12 674 of 14 392) of non-Black patients with mGFR <60 mL min⁻¹ (1.73 m²)⁻¹ (Table 3). Switching to the 2021 eGFR_{Cr}(AS) equation would reclassify an additional 4.7% of Black patients as having an eGFR <60 mL min⁻¹ (1.73 m²)⁻¹. However, 5.9% of Black patients with mGFR >60 mL min⁻¹ (1.73 m²)⁻¹ would

Table 1. Demographic and kidney function characteristics of patients with a clinically ordered iothalamate clearance.

Characteristic	Black	Non-Black	P value
N (%)	852 (2.5%)	33 037 (97.5%)	
Age, y	50.6 $\pm$ 13.9	55.5 $\pm$ 14.0	<0.0001
Age group, N (%)			
<40 y	195 (22.9%)	4891 (14.8%)	<0.0001
40–65 y	533 (62.6%)	19 949 (60.4%)	0.20
>65 y	124 (14.5%)	8197 (24.8%)	<0.0001
BMI, kg/m ²	28.5 $\pm$ 6.1	28.6 $\pm$ 6.0	0.75
BMI category			
<20 kg/m ²	50 (6.5%)	1192 (3.9%)	0.0005
20–24 kg/m ²	165 (21.4%)	7753 (25.4%)	0.0052
25–30 kg/m ²	272 (35.3%)	10 767 (35.3%)	0.68
>30 kg/m ²	283 (36.8%)	10 789 (35.4%)	0.73
Measured GFR	70.5 $\pm$ 35.2	67.2 $\pm$ 32.0	0.01
Measured GFR group-			
<15	39 (4.6%)	943 (2.8%)	0.003
15–29	81 (9.5%)	3376 (10.2%)	0.5
30 to 44	83 (9.7%)	4546 (13.7%)	0.0007
45 to 59	139 (16.3%)	5527 (16.7%)	0.75
60 to 59	260 (30.5%)	10 327 (31.3%)	0.64
≥ 90	250 (29.3%)	8318 (25.2%)	0.006
Serum creatinine	1.69 $\pm$ 1.44	1.40 $\pm$ 0.90	<0.0001

BMI, body-mass index.



be discordantly reclassified for a net reclassification improvement of -1.2% (95CI -5.7 to 3.2). Similar net reclassifications were observed for the 2009 $eGFR_{cr}(ASR-NB)$ equation (Table 3).

At the clinical threshold for transplant eligibility, the 2009 $eGFR_{cr}(ASR)$ equation concordantly classified 67% (50 of 75) of Black patients and 72% (1396 of 1936) of non-Black patients with $mGFR < 20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$. Applying the 2021 $eGFR_{cr}(AS)$ equation correctly reclassified 6.7% ($n=5$) and incorrectly reclassified 0.26% ($n=2$) of Black patients as $< 20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$ (Table 3). No non-Black patients were reclassified to $< 20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$. However, 0.5% ($n=152$) of non-Black patients were correctly reclassified as $> 20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$ while 5.6% ($n=108$) were incorrectly reclassified (Table 3). The net reclassification improvement for the 2021 $eGFR_{cr}(AS)$ equation at $20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$ was 6.4% (95CI 0.41 to 12.4) for Black and -5.1% (95CI -6.0 to -4.1) for non-Black patients.

Discussion

In this report, we compared the new 2021 CKD-EPI $eGFR_{cr}$ equation to the previous 2009 CKD-EPI $eGFR_{cr}$ equation in a large population of patients with and without kidney disease who underwent mGFR for a clinical indication. Our findings confirm the overall accuracy of this new equation (18). At a cohort level, the changes observed in bias, P30, and clinical classification were marginal, and likely clinically insignificant. Our data also confirm that use of the 2009 CKD-EPI $eGFR_{cr}(ASR-NB)$ that did not include the Black coefficient would result in a significantly lower P30 and worse agreement with mGFR clinical categories among Black patients. It should be noted that this equation simply removes the race coefficient and instead uses the non-Black 2009 CKD-EPI $eGFR_{cr}$ equation for all.

One of the goals for the new 2021 CKD-EPI $eGFR_{cr}(AS)$ was to develop one equation for the overall US population, and not report 2 separate values. This approach better represents the diversity present across all patients and within the social constructs of race. Our real-world results suggest that the refit equation has led to an increased underestimation of measured GFR [$-6.06 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$ 2021 $eGFR_{cr}(AS)$ vs $-1.32 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$ 2009 $eGFR_{cr}(ASR)$] among Black people. However, despite the statistically significant changes in bias, our data confirm that the newer 2021 CKD-EPI $eGFR_{cr}(AS)$ equation does not significantly alter the P30 or agreement with mGFR CKD categories.

Two important thresholds of eGFR are 60 and $20 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$. At an eGFR of $60 \text{ mL min}^{-1} (1.73 \text{ m}^2)^{-1}$, Kidney Disease Improving Global

Table 2. Accuracy of estimated GFR (eGFR) compared to measured GFR (mGFR) equations by race groups.

	2009 eGFR _{cr} (ASR)	2009 eGFR _{cr} (ASR-NB)	P value*	2021 eGFR _{cr} (AS)	P value*
Bias: median difference; mL min ⁻¹ (1.73 m ²) ⁻¹ (95CI)					
Black	-1.32 (-2.46 to -0.26)	-8.81 (-9.93 to -7.58)	<0.0001	-6.08 (-7.18 to -4.92)	<0.0001
Non-Black	-3.09 (-3.17 to -3.03)	N/A	-	-0.15 (-0.84 to -0.08)	<0.0001
P30; % (95CI)					
Black	71.7 (68.7 to 74.7)	64.9 (61.7 to 68.1)	0.0025	68.7 (65.6 to 71.8)	0.17
Non-Black	78.6 (78.1 to 79)	N/A	-	78.7 (78.3 to 79.2)	0.63
Agreement; % (95CI)					
Black	55.8 (52.4 to 59.1)	46.2 (42.9 to 49.6)	<0.0001	51.4 (48.1 to 54.8)	0.072
Non-Black	55.0 (54.5 to 55.6)	N/A	-	57.1 (56.5 to 57.6)	<0.0001

*P value compared with 2009 eGFR_{cr}(ASR).**Table 3.** Reclassification of patients into correct measured GFR clinical thresholds.

	Black		Non-Black
	2009 eGFR _{cr} (ASR-NB)	2021 eGFR _{cr} (AS)	2021 eGFR _{cr} (AS)
CKD reclassification*			
<60 mL min ⁻¹ (1.73 m ²) ⁻¹	8.8% (5.7 to 11.8)	4.7% (2.3 to 7.0)	-4.1% (-3.8 to -4.4)
>60 mL min ⁻¹ (1.73 m ²) ⁻¹	-12.6% (-15.4 to -9.7)	-5.9% (-8.0 to -3.8)	4.4% (4.1 to 4.7)
Net	-3.8% (-9.7 to 2.1)	-1.2% (-5.7 to 3.2)	0.33% (-0.3 to 0.9)
Kidney failure reclassification*			
<20 mL min ⁻¹ (1.73 m ²) ⁻¹	8.0% (1.35 to 14.7)	6.7% (1.02 to 12.3)	-5.6% (-4.5 to -6.6)
>20 mL min ⁻¹ (1.73 m ²) ⁻¹	-1.03 (-1.82 to -0.24)	-0.26% (-0.61 to 0.10)	0.49% (0.41 to 0.57)
Net	7.0% (-0.53 to 14.5)	6.4% (0.36 to 12.4)	-5.1% (-6.0 to -4.1)

*Relative increase or decrease in mGFR agreement compared to 2009 eGFR_{cr}(ASR). Parenthesis indicate 95th percentile confidence intervals. Confidence intervals that include 0 are not significantly different than classification agreement using 2009eGFR_{cr}(ASR).

Outcomes guidelines recommend workup and treatment for chronic kidney disease. At a threshold of <20 mL min⁻¹ (1.73 m²)⁻¹, patients are considered to have incipient kidney failure and, in the USA, are eligible for deceased-donor kidney transplantation listing. At the CKD 3A eGFR threshold, almost equal numbers in both race groups were reclassified by the 2021 eGFR_{cr}(AS) as above or below 60 mL min⁻¹ (1.73 m²)⁻¹. Thus, in this cohort the net reclassification for CKD diagnosis was not statistically different than zero when using the 2021 eGFR_{cr}(AS) equation compared to 2009 eGFR_{cr}(ASR). At the kidney failure cut point for transplant listing, an additional 6.7% of Black patients would correctly fall below 20 mL min⁻¹ (1.73 m²)⁻¹ when using the 2021

eGFR_{cr}(AS) compared to the 2009 eGFR_{cr}(ASR) equation. The reverse was true for non-Black patients, which would see a net decrease of 5.1% in patients correctly classified as <20 mL min⁻¹ (1.73 m²)⁻¹. Both results are as expected from the relative effect of refitting the 2021 eGFR_{cr}(AS) equation in the racial groups at these thresholds (18). It should be noted that despite the significant adjustments to reclassification, a large number of patients remain incorrectly classified due to the inherent uncertainty in eGFR (7, 16).

Overall, our results confirm that there would be relatively minimal effect on reported eGFR [± 5 mL min⁻¹ (1.73 m²)⁻¹] in the population at large. However, at lower eGFR, particularly near critical decision thresholds, these changes may lead to a relatively large number

of individuals that would change staging once the newer 2021 CKD-EPI $eGFR_{Cr}(AS)$ equation is used as compared to the 2009 CKD-EPI $eGFR_{Cr}(ASR)$ version. On an individual patient level, these results also highlight the importance of confirming creatinine-based $eGFR$ when results fall near an important action point, either using an alternative biomarker (such as Cystatin C) or directly measuring GFR.

Our study has several limitations. The cohort was evaluated at a single academic medical center and the observed P30 and agreement were lower than what was reported in the CKD-EPI cross-validation (18). The overall percentage of Black subjects in our data was relatively low (2.5%). The imbalance between Black and non-Black patients in our study cohort prevents direct comparison of equation performance between race groups. However, given the long time frame available for analysis, the absolute number of Black patients ($n = 852$) was sufficient for meaningful analysis. In addition, while iothalamate clearance was measured using a standardized protocol and serum creatinine was measured by a standardized enzymatic assay, studies have demonstrated that imprecision in gold-standard measures of GFR contribute to an appreciable proportion of cases in which $eGFR$ and $mGFR$ differ by $\geq 30\%$ (16).

Healthcare has adjusted to a more rapid pace of change. Many laboratories with good intentions have already implemented the 2009 $eGFR_{Cr}(ASR-NB)$. Rapid and uniform implementation of the new 2021

$eGFR_{Cr}(AS)$ equation will preserve the gains made after many decades of work toward standardized creatinine measures, $eGFR$ reporting, and indirectly improve equitable access to CKD resources.

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References

- Levey AS, Bosch JP, Lewis JB, Greene T, Rogers N, Roth D. A more accurate method to estimate glomerular filtration rate from serum creatinine: a new prediction equation. Modification of diet in renal disease study group. *Ann Intern Med* 1999;130:461-70.
- Vyas DA, Eisenstein LG, Jones DS. Hidden in plain sight - reconsidering the use of race correction in clinical algorithms. *N Engl J Med* 2020;383:874-82.
- Hogarth RA. Medicalizing blackness: making racial differences in the Atlantic world, 1780-1840. Chapel Hill (NC): The University of North Carolina Press; 2017. p. 290.
- Washington HA. Medical apartheid: the dark history of medical experimentation on Black Americans from colonial times to the present. 1st Ed. New York (NY): Doubleday; 2006. p. 501.
- Feagin J, Bennefield Z. Systemic racism and U.S. Health care. *Soc Sci Med* 2014;103:7-14.
- Levey AS, Stevens LA, Schmid CH, Zhang YL, Castro AF, Feldman HI, et al.; for the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration). A new equation to estimate glomerular filtration rate. *Ann Intern Med* 2009;150:604-12.
- Miller WG. Uncertainty in estimated glomerular filtration rate is much larger than the race adjustment term. *Clin Chem* 2021;67:693-5.
- Yudell M, Roberts D, DeSalle R, Tishkoff S. Science and society. Taking race out of human genetics. *Science* 2016;351:564-5.
- Fontanarosa PB, Bauchner H. Race, ancestry, and medical research. *JAMA* 2018;320:1539-40.
- Nelson A. Unequal treatment: confronting racial and ethnic disparities in health care. *J Natl Med Assoc* 2002;94:666-8.
- Quinn MP, Rainey A, Cairns KJ, Marshall AH, Savage G, Kee F, et al. The practical implications of using standardized estimation equations in calculating the prevalence of chronic kidney disease. *Nephrol Dial Transplant* 2007;23:542-8.
- Miller WG, Jones GRD. Estimated glomerular filtration rate; laboratory implementation and current global status. *Adv Chronic Kidney Dis* 2018;25:7-13.
- Stevens LA, Coresh J, Greene T, Levey AS. Assessing kidney function-measured and estimated glomerular filtration rate. *N Engl J Med* 2006;354:2473-83.
- Levey AS, Coresh J, Tighiouart H, Greene T, Inker LA. Measured and estimated glomerular filtration rate: current status and future directions. *Nat Rev Nephrol* 2020;16:51-64.
- Eneanya ND, Yang W, Reese PP. Reconsidering the consequences of using race to estimate kidney function. *JAMA* 2019;322:113-4.
- Kwong YT, Stevens LA, Selvin E, Zhang YL, Greene T, Van Lente F, et al. Imprecision of urinary iothalamate clearance as a gold-standard measure of GFR decreases the diagnostic accuracy of kidney function estimating equations. *Am J Kidney Dis* 2010;56:39-49.
- Ahmed S, Nutt CT, Eneanya ND, Reese PP, Sivashanker K, Morse M, et al. Examining the potential impact of race multiplier utilization in estimated glomerular filtration rate calculation on African-American care outcomes. *J Gen Intern Med* 2021;36:464-71.
- Inker LA, Eneanya ND, Coresh J, Tighiouart H, Wang D, Sang Y, et al. New creatinine- and cystatin C-based equations to estimate GFR without race. *N Engl J Med* 2021;385:1737-49.
- Delgado C, Baweja M, Crews D, Eneanya N, Gadegebeku C, Inker L, et al. A unifying approach for GFR estimation: recommendations of the NKF-ASN task force on reassessing the inclusion of race in diagnosing kidney disease. *J Am Soc Nephrol* 2021;32:2994-3015.
- Seegmiller JC, Burns BE, Fauq AH, Mukhtar N, Lieske JC, Larson TS. Iothalamate quantification by tandem mass spectrometry to measure glomerular filtration rate. *Clin Chem* 2010;56:568-74.
- Du BD, Du BE. A formula to estimate the approximate surface area if height and weight be known. *Arch Intern Med (Chic)* 1916;XVII:863-71.
- Pieroni L, Delanaye P, Boutten A, Bargnoux AS, Rozet E, Delatour V, et al.; Société Française de Biologie Clinique. A multicentric evaluation of IDMS-traceable creatinine enzymatic assays. *Clin Chim Acta* 2011;412:2070-5.